Lecture 23: Ideal Observer

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🔑 Learning Objectives

By the end of this lecture, you should be able to understand:

- Ideal observer (IO) with independent, identically distributed (i.i.d.) Gaussian noise
- IO with correlated Gaussian noise
- SNR and AUC in the Gaussian case

Ideal Observer (Bayesian Observer)

The **ideal observer (IO)** utilizes *all available statistical information* about the task to maximize performance, typically measured by **Bayes risk** or a related metric.

The performance of IO provides a theoretical *upper bound* against which all practical observers can be compared.

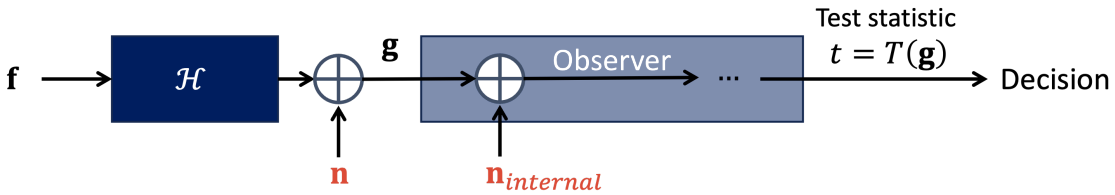
The IO applies a **likelihood-ratio test** to distinguish between hypotheses:

$$\text{Choose } H_2 \text{ if } \Lambda(\mathbf{g}) > \Lambda_c, \quad \text{where } \Lambda(\mathbf{g}) = \frac{\text{pdf}(\mathbf{g}|H_2)}{\text{pdf}(\mathbf{g}|H_1)}$$

Requirements for an IO:

- **No internal noise**: The IO does not introduce additional uncertainty, internal noise mechanism that would further corrupt the data.
- **Known threshold**: The decision threshold Λ_c is known and consistent across trials.

- **Complete knowledge**: The IO must have access to the conditional probability density functions (PDFs) $\text{pdf}(\mathbf{g}|H_j)$.



The choice of threshold Λ_c determines the decision boundary that yields **minimum Bayes risk** or **minimum-error** performance.

The IO achieves:

- Maximum TPF for any FPF
- Maximum AUC of all observers

IO with Independent, Identically Distributed Gaussian Noise

To illustrate the theory, consider the **signal-known-exactly/background-known-exactly (SKE/BKE)** problem. This is an *idealized scenario* in which both the signal and background are deterministic and completely known.

Hypotheses

$H_1 : \mathbf{g} = \mathcal{H}\mathbf{f}_1 + \mathbf{n} = \mathbf{s}_1 + \mathbf{n}$
$H_2 : \mathbf{g} = \mathcal{H}\mathbf{f}_2 + \mathbf{n} = \mathbf{s}_2 + \mathbf{n}$

- $\mathbf{f}_j \in \mathbb{R}^N$: Known object vector (nonrandom)
- $\mathbf{s}_j \in \mathbb{R}^M$: Data space vector resulting from the imaging operator $\mathcal{H}$ acting on the object space vector $\mathbf{f}_j$
- $\mathbf{n}$: i.i.d. Gaussian noise (random) $n_m \sim \mathcal{N}(0, \sigma^2)$

Each data component g_m is corrupted by an independent Gaussian noise term $n_m \sim \mathcal{N}(0, \sigma^2)$. Thus, the variance σ^2 is consistent across all data elements.

To compute the likelihood ratio $\Lambda(\mathbf{g}) = \frac{\text{pdf}(\mathbf{g}|H_2)}{\text{pdf}(\mathbf{g}|H_1)}$, we should first compute the conditional PDF $\text{pdf}(\mathbf{g}|H_j)$.

Under hypothesis H_j, each component g_m follows a Gaussian distribution:

$$\begin{aligned}\text{pdf}(g_m|H_j) &= \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-\frac{(g_m-\mathrm{E}[g_m])^2}{2\sigma^2}\right]\\ &= \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-\frac{(g_m-s_{jm})^2}{2\sigma^2}\right]\end{aligned}$$

Assuming the components of $\mathbf{g}$ are *statistically independent*, the joint conditional density is:

$$\begin{aligned}\text{pdf}(\mathbf{g}|H_j) &= \prod_{m=1}^{M}\text{pdf}(g_m|H_j)\\ &= \prod_{m=1}^{M}\left[\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left[-\frac{(g_m-s_{jm})^2}{2\sigma^2}\right]\right]\\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^M\prod_{m=1}^{M}\exp\left[-\frac{(g_m-s_{jm})^2}{2\sigma^2}\right]\end{aligned}$$

Likelihood Ratio

$$\Lambda(\mathbf{g}) = \frac{\text{pdf}(\mathbf{g}|H_2)}{\text{pdf}(\mathbf{g}|H_1)} = \frac{\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^M\prod_{m=1}^{M}\exp\left[-\frac{(g_m-s_{2m})^2}{2\sigma^2}\right]}{\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^M\prod_{m=1}^{M}\exp\left[-\frac{(g_m-s_{1m})^2}{2\sigma^2}\right]}$$

$$= \frac{\prod_{m=1}^{M}\exp\left[-\frac{(g_m-s_{2m})^2}{2\sigma^2}\right]}{\prod_{m=1}^{M}\exp\left[-\frac{(g_m-s_{1m})^2}{2\sigma^2}\right]} = \prod_{m=1}^{M}\exp\left[\frac{(g_m-s_{1m})^2}{2\sigma^2}-\frac{(g_m-s_{2m})^2}{2\sigma^2}\right]$$

$$= \prod_{m=1}^{M}\exp\left[\frac{g_m^2-2g_ms_{1m}+s_{1m}^2}{2\sigma^2}-\frac{g_m^2-2g_ms_{2m}+s_{2m}^2}{2\sigma^2}\right]$$

$$= \prod_{m=1}^{M}\exp\left[\frac{2g_ms_{2m}-2g_ms_{1m}}{2\sigma^2}-\frac{s_{2m}^2-s_{1m}^2}{2\sigma^2}\right]$$

$$= \prod_{m=1}^{M}\exp\left[\frac{(s_{2m}-s_{1m})g_m}{\sigma^2}-\frac{s_{2m}^2-s_{1m}^2}{2\sigma^2}\right]\underset{D_1}{\overset{D_2}{\gtrless}}\Lambda_c$$

Log-Likelihood Ratio

Taking logarithms:

$$\lambda(\mathbf{g}) = \ln \Lambda(\mathbf{g}) = \ln \prod_{m=1}^{M} \exp\left[\frac{(s_{2m} - s_{1m})g_m}{\sigma^2} - \frac{s_{2m}^2 - s_{1m}^2}{2\sigma^2}\right]$$

$$= \sum_{m=1}^{M}\left[\frac{(s_{2m} - s_{1m})g_m}{\sigma^2} - \frac{s_{2m}^2 - s_{1m}^2}{2\sigma^2}\right] \underset{D_1}{\overset{D_2}{\gtrless}} \ln \Lambda_c = \lambda_c$$

Since the second term in each summand $\frac{s_{2m}^2 - s_{1m}^2}{2\sigma^2}$ is **independent of the data $\mathbf{g}$**, we can define a simplified test statistic:

$$\lambda'(\mathbf{g}) = \sum_{m=1}^{M}(s_{2m} - s_{1m})g_m \underset{D_1}{\overset{D_2}{\gtrless}} \lambda_c'$$

$$\lambda'(\mathbf{g}) = \Delta\mathbf{s}^T\mathbf{g} \underset{D_1}{\overset{D_2}{\gtrless}} \lambda_c', \quad \Delta\mathbf{s} = \mathbf{s}_2 - \mathbf{s}_1$$

This expression defines the well-known **matched-filter** discriminator, where the expected signal difference $\Delta\mathbf{s} = \mathbf{s}_2 - \mathbf{s}_1$ serves as a filter that is correlated with the data $\mathbf{g}$.

Matched filtering is widely recognized as the **optimal detection strategy** for known signals embedded in additive white Gaussian noise.

IO with Correlated Gaussian Noise

When the additive noise $\mathbf{n}$ is correlated, the covariance matrix under each hypothesis is:

$$\mathbf{K}_j = \mathrm{E}[(\mathbf{g} - \mathbf{s}_j)(\mathbf{g} - \mathbf{s}_j)^T | H_j] = \mathrm{E}[\mathbf{n}\mathbf{n}^T]$$

Since the noise is independent of the signal:

$$\mathbf{K}_1 = \mathbf{K}_2 = \mathbf{K}_\mathbf{n}$$

Here, each noise component $n_m \sim \mathcal{N}(0, \sigma_m^2)$, the variance and unlike the i.i.d. case, σ_m may vary with m, requiring the use of the **full covariance matrix $\mathbf{K}_\mathbf{n}$**.

Under hypothesis H_j, the measurement vector $\mathbf{g}$ follows a multivariate Gaussian distribution:

$$\mathrm{pdf}(\mathbf{g}|H_j) = \frac{1}{\sqrt{(2\pi)^M \det(\mathbf{K}_\mathbf{n})}} \exp\left[-\frac{1}{2}(\mathbf{g} - \mathbf{s}_j)^T\mathbf{K}_\mathbf{n}^{-1}(\mathbf{g} - \mathbf{s}_j)\right]$$

Likelihood Ratio

$$\Lambda(\mathbf{g}) = \frac{\text{pdf}(\mathbf{g}|H_2)}{\text{pdf}(\mathbf{g}|H_1)}$$
$$= \frac{\frac{1}{\sqrt{(2\pi)^M \det(\mathbf{K_n})}} \exp\left[-\frac{1}{2}(\mathbf{g}-\mathbf{s}_2)^T\mathbf{K_n}^{-1}(\mathbf{g}-\mathbf{s}_2)\right]}{\frac{1}{\sqrt{(2\pi)^M \det(\mathbf{K_n})}} \exp\left[-\frac{1}{2}(\mathbf{g}-\mathbf{s}_1)^T\mathbf{K_n}^{-1}(\mathbf{g}-\mathbf{s}_1)\right]}$$
$$= \frac{\exp\left[-\frac{1}{2}(\mathbf{g}-\mathbf{s}_2)^T\mathbf{K_n}^{-1}(\mathbf{g}-\mathbf{s}_2)\right]}{\exp\left[-\frac{1}{2}(\mathbf{g}-\mathbf{s}_1)^T\mathbf{K_n}^{-1}(\mathbf{g}-\mathbf{s}_1)\right]} \underset{D_1}{\overset{D_2}{\gtrless}} \Lambda_c$$

Log-Likelihood Ratio

Taking logarithms:

$$\lambda(\mathbf{g}) = \ln \Lambda(\mathbf{g})$$
$$= \ln \exp\left[\frac{1}{2}(\mathbf{g}-\mathbf{s}_1)^T\mathbf{K_n}^{-1}(\mathbf{g}-\mathbf{s}_1) - \frac{1}{2}(\mathbf{g}-\mathbf{s}_2)^T\mathbf{K_n}^{-1}(\mathbf{g}-\mathbf{s}_2)\right]$$
$$= \frac{1}{2}(\mathbf{g}-\mathbf{s}_1)^T\mathbf{K_n}^{-1}(\mathbf{g}-\mathbf{s}_1) - \frac{1}{2}(\mathbf{g}-\mathbf{s}_2)^T\mathbf{K_n}^{-1}(\mathbf{g}-\mathbf{s}_2)$$
$$= \frac{1}{2}\left[(\mathbf{g}-\mathbf{s}_1)^T\mathbf{K_n}^{-1}(\mathbf{g}-\mathbf{s}_1) - \mathbf{g}-\mathbf{s}_2)^T\mathbf{K_n}^{-1}(\mathbf{g}-\mathbf{s}_2)\right]$$
$$= \frac{1}{2}\left[\mathbf{g}^T\mathbf{K_n}^{-1}\mathbf{g} - \mathbf{s}_1^T\mathbf{K_n}^{-1}\mathbf{g} - \mathbf{g}^T\mathbf{K_n}^{-1}\mathbf{s}_1 + \mathbf{s}_1^T\mathbf{K_n}^{-1}\mathbf{s}_1 - \mathbf{g}^T\mathbf{K_n}^{-1}\mathbf{g} + \mathbf{s}_2^T\mathbf{K_n}^{-1}\mathbf{g} + \mathbf{g}^T\mathbf{K_n}^{-1}\mathbf{s}_2 - \mathbf{s}_2^T\mathbf{K_n}^{-1}\mathbf{s}_2\right]$$

$$= \frac{1}{2}\left[-\mathbf{s}_1^T\mathbf{K_n}^{-1}\mathbf{g} - \mathbf{g}^T\mathbf{K_n}^{-1}\mathbf{s}_1 + \mathbf{s}_1^T\mathbf{K_n}^{-1}\mathbf{s}_1 + \mathbf{s}_2^T\mathbf{K_n}^{-1}\mathbf{g} + \mathbf{g}^T\mathbf{K_n}^{-1}\mathbf{s}_2 - \mathbf{s}_2^T\mathbf{K_n}^{-1}\mathbf{s}_2\right]$$
$$= \frac{1}{2}\left[(\mathbf{s}_2-\mathbf{s}_1)^T\mathbf{K_n}^{-1}\mathbf{g} + \mathbf{g}^T\mathbf{K_n}^{-1}(\mathbf{s}_2-\mathbf{s}_1) + \mathbf{s}_1^T\mathbf{K_n}^{-1}\mathbf{s}_1 - \mathbf{s}_2^T\mathbf{K_n}^{-1}\mathbf{s}_2\right]$$
$$= \frac{1}{2}\Big[\underbrace{\Delta\mathbf{s}^T}_{(1\times M)}\underbrace{\mathbf{K_n}^{-1}}_{(M\times M)}\underbrace{\mathbf{g}}_{(M\times 1)} + \underbrace{\mathbf{g}^T}_{(1\times M)}\underbrace{\mathbf{K_n}^{-1}}_{(M\times M)}\underbrace{\Delta\mathbf{s}}_{(M\times 1)} + \mathbf{s}_1^T\mathbf{K_n}^{-1}\mathbf{s}_1 - \mathbf{s}_2^T\mathbf{K_n}^{-1}\mathbf{s}_2\Big]$$

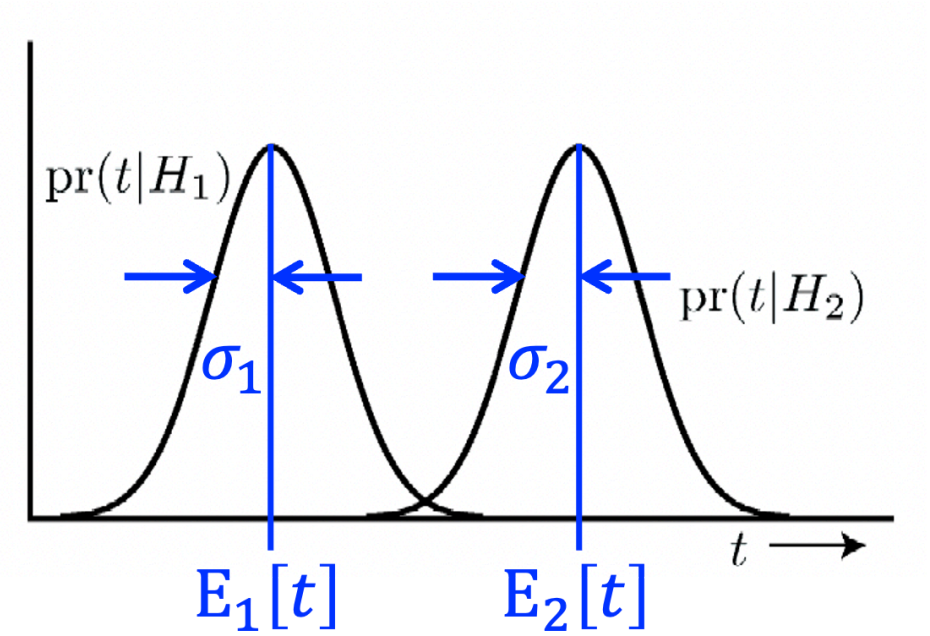
The last two terms are *constants (scalars) independent of the data* $\mathbf{g}$. Since the covariance matrix $\mathbf{K_n}$ is symmetric, the first term can be rewritten as:

$$\lambda'(\mathbf{g}) = \Delta\mathbf{s}^T\mathbf{K_n}^{-1}\mathbf{g} \underset{D_1}{\overset{D_2}{\gtrless}} \lambda'_c$$

This defines a linear test statistic analogous to the matched filter, but adapted for the case of *correlated* Gaussian noise.

SNR and AUC in the Gaussian Case

Assume the data $\mathbf{g}$ are multivariate normal and that the noise covariance is the same under both hypotheses:



Foundations of image science, Barrett and Myers

$$\mathbf{K}_1 = \mathbf{K}_2 = \mathbf{K}_\mathbf{n}.$$

The decision variable $\lambda'(\mathbf{g})$ is a linear transformation of the data vector $\mathbf{g}$. Since linear transformations of Gaussian random variables remain Gaussian, it follows that $\lambda'(\mathbf{g})$ is also Gaussian.

To compute the SNR of the ideal observer's test statistic $\lambda'(\mathbf{g})$, we must find its mean and variance under each hypothesis.

Mean of the Decision Variable

$$\begin{aligned}\mathrm{E}[\lambda(\mathbf{g})|H_j] &= \mathrm{E}_j[\lambda(\mathbf{g})] = \mathrm{E}_j[\Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{g}] \\ &= \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathrm{E}_j[\mathbf{g}] = \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathrm{E}[\mathbf{s}_j + \mathbf{n}] \\ &= \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\big(\mathbf{s}_j + \mathrm{E}[\mathbf{n}]\big) = \underbrace{\Delta\mathbf{s}^T}_{(1\times M)}\underbrace{\mathbf{K}_\mathbf{n}^{-1}}_{(M\times M)}\underbrace{\mathbf{s}_j}_{(M\times 1)}\end{aligned}$$

Variance of the Decision Variable

$$\begin{aligned}&\sigma_\lambda^2 = \mathrm{E}\big[\big(\lambda(\mathbf{g}) - \mathrm{E}[\lambda(\mathbf{g})|H_j]\big)^2|H_j\big] \\ &= \mathrm{E}_j\big[\big(\lambda(\mathbf{g}) - \mathrm{E}_j[\lambda(\mathbf{g})]\big)^2\big] \\ &= \mathrm{E}_j\big[\lambda(\mathbf{g})^2 - 2\lambda(\mathbf{g})\mathrm{E}_j[\lambda(\mathbf{g})] + \mathrm{E}_j[\lambda(\mathbf{g})]^2\big] \\ &= \mathrm{E}_j\big[\lambda(\mathbf{g})^2\big] - 2\mathrm{E}_j\big[\lambda(\mathbf{g})\mathrm{E}_j[\lambda(\mathbf{g})]\big] + \mathrm{E}_j\big[\mathrm{E}_j[\lambda(\mathbf{g})]^2\big] \\ &= \mathrm{E}_j[\lambda(\mathbf{g})^2] - 2\mathrm{E}_j[\lambda(\mathbf{g})]^2 + \mathrm{E}_j[\lambda(\mathbf{g})]^2 \\ &= \mathrm{E}_j[\lambda(\mathbf{g})^2] - \mathrm{E}_j[\lambda(\mathbf{g})]^2\end{aligned}$$

$$
\begin{aligned}
&= \mathrm{E}_j[(\Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{g})^2] - (\Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{s}_j)^2 \quad \text{(scalar)} \\
&= \mathrm{E}_j[(\Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{g})(\Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{g})^T] \\
&- (\Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{s}_j)(\Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{s}_j)^T \quad \text{(Hermitian)} \\
&= \mathrm{E}_j[\Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{g}\mathbf{g}^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s}] - \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{s}_j\mathbf{s}_j^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} \\
&= \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathrm{E}_j[\mathbf{g}\mathbf{g}^T]\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} - \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{s}_j\mathbf{s}_j^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} \\
&= \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathrm{E}[(\mathbf{s}_j + \mathbf{n})(\mathbf{s}_\mathbf{j} + \mathbf{n})^T]\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} - \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{s}_j\mathbf{s}_j^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} \\
&= \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathrm{E}[(\mathbf{s}_j + \mathbf{n})(\mathbf{s}_j^T + \mathbf{n}^T)]\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} - \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{s}_j\mathbf{s}_j^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s}
\end{aligned}
$$

$$
\begin{aligned}
&= \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathrm{E}[\mathbf{s}_j\mathbf{s}_j^T + \mathbf{s}_j^T\mathbf{n} + \mathbf{s}_j\mathbf{n}^T + \mathbf{n}\mathbf{n}^T]\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} - \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{s}_j\mathbf{s}_j^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} \\
&= \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\big(\mathbf{s}_j\mathbf{s}_j^T + \mathbf{s}_j^T\mathrm{E}[\mathbf{n}] + \mathbf{s}_j\mathrm{E}[\mathbf{n}^T] + \mathrm{E}[\mathbf{n}\mathbf{n}^T]\big)\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} - \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{s}_j\mathbf{s}_j^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} \\
&= \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\big(\mathbf{s}_j\mathbf{s}_j^T + \mathrm{E}[\mathbf{n}\mathbf{n}^T]\big)\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} - \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{s}_j\mathbf{s}_j^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} \\
&= \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\big(\mathbf{s}_j\mathbf{s}_j^T + \mathbf{K}_\mathbf{n}\big)\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} - \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{s}_j\mathbf{s}_j^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} \\
&= \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{s}_j\mathbf{s}_j^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} + \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{K}_\mathbf{n}\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} - \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{s}_j\mathbf{s}_j^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s} \\
&= \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s}
\end{aligned}
$$

SNR of the Decision Variable

$$
\mathrm{SNR}_\lambda = \frac{\mathrm{E}_2[\lambda(\mathbf{g})] - \mathrm{E}_1[\lambda(\mathbf{g})]}{\sqrt{\frac{1}{2}\sigma_\lambda^2 + \frac{1}{2}\sigma_\lambda^2}}
$$

$$
\mathrm{SNR}_\lambda^2 = \frac{\big(\mathrm{E}_2[\lambda(\mathbf{g})] - \mathrm{E}_1[\lambda(\mathbf{g})]\big)^2}{\frac{1}{2}\sigma_\lambda^2 + \frac{1}{2}\sigma_\lambda^2} = \frac{\big(\mathrm{E}_2[\lambda(\mathbf{g})] - \mathrm{E}_1[\lambda(\mathbf{g})]\big)^2}{\sigma_\lambda^2}
$$

$$
= \frac{\big(\Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{s}_2 - \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\mathbf{s}_1\big)^2}{\Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s}} = \frac{\big(\Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s}\big)^2}{\Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s}} = \Delta\mathbf{s}^T\mathbf{K}_\mathbf{n}^{-1}\Delta\mathbf{s}
$$

- Uncorrelated noise with variance σ_m^2:

$$
\mathrm{SNR}_\lambda^2 = \sum_{m=1}^{M} \frac{(\Delta\mathbf{s}_m)^2}{\sigma_m^2}
$$

- White noise, uncorrelated noise with uniform variance ($\sigma_m = \sigma$)

$$
\mathrm{SNR}_\lambda^2 = \sum_{m=1}^{M} \frac{(\Delta\mathbf{s}_m)^2}{\sigma^2} = \frac{1}{\sigma^2}\sum_{m=1}^{M}(\Delta\mathbf{s}_m)^2 = \frac{\|\Delta\mathbf{s}\|^2}{\sigma^2}
$$

Prewhitening Matched Filter

If the noise covariance matrix $\mathbf{K}_\mathbf{n}$ is nonsingular, we can transform the data by applying the square-root inverse of the covariance matrix to decorrelate the noise:

$$\mathbf{z} = \mathbf{K_n}^{-\frac{1}{2}}\mathbf{g}$$

Mean

$$\mathrm{E}[\mathbf{z}|H_j] = \mathrm{E}_j[\mathbf{z}] = \mathbf{K_n}^{-\frac{1}{2}}\mathrm{E}_j[\mathbf{g}]$$
$$\Delta\mathrm{E}[\mathbf{z}] = \mathrm{E}_2[\mathbf{z}] - \mathrm{E}_1[\mathbf{z}] = \mathbf{K_n}^{-\frac{1}{2}}\big(\mathrm{E}_2[\mathbf{g}] - \mathrm{E}_1[\mathbf{g}]\big) = \mathbf{K_n}^{-\frac{1}{2}}\Delta\mathbf{s}$$

Test Statistic

The test statistic from the log-likelihood ratio becomes:

$$\lambda = \Delta\mathbf{s}^T\mathbf{K_n}^{-1}\mathbf{g} = \Delta\mathbf{s}^T\mathbf{K_n}^{-\frac{1}{2}}\mathbf{K_n}^{-\frac{1}{2}}\mathbf{g} = \big[\mathbf{K_n}^{-\frac{1}{2}}\Delta\mathbf{s}\big]^T\big[\mathbf{K_n}^{-\frac{1}{2}}\mathbf{g}\big]$$
$$= \big(\mathrm{E}_2[\mathbf{z}] - \mathrm{E}_1[\mathbf{z}]\big)^T\mathbf{z} = \Delta\mathrm{E}[\mathbf{z}]^T\mathbf{z}$$

Covariance

The covariance of $\mathbf{z}$ is:

$$\mathbf{K_z} = \mathrm{E}\big[\Delta\mathbf{z}\Delta\mathbf{z}^T\big] = \mathrm{E}\big[\big(\mathbf{K_n}^{-\frac{1}{2}}\Delta\mathbf{g}\big)\big(\mathbf{K_n}^{-\frac{1}{2}}\Delta\mathbf{g}\big)^T\big] = \mathrm{E}\big[\mathbf{K_n}^{-\frac{1}{2}}\Delta\mathbf{g}\Delta\mathbf{g}^T\mathbf{K_n}^{-\frac{1}{2}}\big]$$
$$= \mathbf{K_n}^{-\frac{1}{2}}\mathrm{E}\big[\Delta\mathbf{g}\Delta\mathbf{g}^T\big]\mathbf{K_n}^{-\frac{1}{2}} = \mathbf{K_n}^{-\frac{1}{2}}\mathbf{K_n}\mathbf{K_n}^{-\frac{1}{2}} = \mathbf{I}$$

This yields a Gaussian random vector with whitened noise, meaning the noise has a flat (white) eigenspectrum. The ideal strategy is to first prewhiten the data, then apply a matched filter to the prewhitened signal difference. This observer is known as the prewhitening matched filter (PWMF).

SNR

$$\mathrm{SNR}_\lambda^2 = \frac{\|\mathrm{E}_2[\mathbf{z}] - \mathrm{E}_1[\mathbf{z}]\|^2}{\sigma^2} = \|\mathrm{E}_2[\mathbf{z}] - \mathrm{E}_1[\mathbf{z}]\|^2 = \|\mathbf{K_n}^{-\frac{1}{2}}\Delta\mathbf{s}\|^2$$